Yingru (Richard) Li

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Reinforcement Learning · LLM · Optimization · Reasoning & Agents

Experience

xAI - SpaceX Singapore
Member of Technical Staff *December 2025 – Present*

- Drive large-scale ("big-run") reinforcement learning for the Grok models (Grok 4.x, Grok Build), spanning both the **science** and the **infrastructure** of RL training.
- **Science of RL** — stability, efficiency, and the core algorithms and recipes; identified and resolved critical training instabilities that benefit every stage of the RL pipeline, directly enabling the longest stable RL run.
- **Long-horizon RL** — developing recipes for higher intelligence per token and recursive self-improvement: RL training with state compaction, self-distillation for dense credit assignment, beyond 1 bit per rollout.
- **RL infrastructure** — built and scaled the systems behind these recipes, including big-run training across $\gg$10,000 GPUs.

ByteDance Singapore
LLM Algorithm Research Scientist *2025 – December 2025*

- Led research on RL training stability and efficiency for LLM reasoning and agents.
- Diagnosed and formalized the **RL Collapse** phenomenon, tracing instability to implicit off-policyness caused by the training–inference mismatch (which induces gradient variance and bias); developed **Trust Region Masking**, **Dynamic Vocabulary Pruning**, the **Optimal Token Baseline**, and a simple LR-scheduling fix — complementary methods that together address these root causes.
- Co-led **SimpleTIR** (end-to-end multi-turn tool-integrated reasoning) and **Dr. Kernel** (Triton kernel generation agent via RL).

Tencent AI & Robotics X Shenzhen, China
Student Research Staff Member, Agent Center *2019 – 2022*

- Led **HyperAgent** (ICML 2024) and **HyperDQN** (ICLR 2022) for scalable exploration in deep RL, and **Divergence-Augmented Policy Optimization** (NeurIPS 2019) for stable off-policy data reuse.

Microsoft Research Lab – Asia Beijing, China
Research Intern, Theory Center *2016*

- Research on combinatorial optimization and reinforcement / bandit learning algorithms.

Selected Publications

Full list at Google Scholar. ⋆ denotes equal contribution; † denotes corresponding author.

1. **The Optimal Token Baseline: Variance Reduction for Long-Horizon LLM-RL**
   Yingru Li, Jiawei Xu, Ziniu Li, Jiacai Liu, Wei Liu, Yuxuan Tong, Longtao Zheng, Zhenghai Xue, Yaxiang Zhang, Tianle Cai, Ge Zhang, Qian Liu, Baoxiang Wang. *ICML 2026.*
2. **Trust Region Masking for Long-Horizon LLM Reinforcement Learning**
   Yingru Li, Jiacai Liu, Jiawei Xu, Yuxuan Tong, Ziniu Li, Qian Liu, Baoxiang Wang. *ICML 2026.*
3. **When Speed Kills Stability: Demystifying RL Collapse from the Training-Inference Mismatch**
   Jiacai Liu⋆, **Yingru Li**⋆,†, Yuqian Fu, Jiawei Wang, Qian Liu, Yu Shen. *Technical Report, 2025.*
4. **Taming the Tail: Stable LLM Reinforcement Learning via Dynamic Vocabulary Pruning**
   Yingru Li, Jiawei Xu, Jiacai Liu, Yuxuan Tong, Ziniu Li, Tianle Cai, Ge Zhang, Qian Liu, Baoxiang Wang. *arXiv 2025.*
5. **Harnessing Uncertainty: Entropy-Modulated Policy Gradients for Long-Horizon LLM Agents**
   Jiawei Wang, Jiacai Liu, Yuqian Fu, **Yingru Li**, Xintao Wang, Yuan Lin, Yu Yue, Lin Zhang, Yang Wang, Ke Wang. *ICML 2026.*
6. **SimpleTIR: End-to-End Reinforcement Learning for Multi-Turn Tool-Integrated Reasoning**
   Zhenghai Xue, Longtao Zheng, Qian Liu, **Yingru Li**, Xiaosen Zheng, Zejun Ma, Bo An. *ICLR 2026.*

7. **Dr. Kernel: Reinforcement Learning Done Right for Triton Kernel Generations**
   Wei Liu, Jiawei Xu, **Yingru Li**, Longtao Zheng, Tianjian Li, Qian Liu, Junxian He. *ICML 2026.*
8. **Owl: Optimized Workforce Learning for General Multi-Agent Assistance in Real-World Task Automation**
   Mengkang Hu, Yuhang Zhou, Wendong Fan, . . . , **Yingru Li**, . . . , Bernard Ghanem, Ping Luo, Guohao Li. *NeurIPS 2025.*
9. **Scalable Exploration via Ensemble++**
   Yingru Li, Jiawei Xu, Baoxiang Wang, Zhi-Quan Luo. *NeurIPS 2025.*
10. **Q-Star Meets Scalable Posterior Sampling: Bridging Theory and Practice via HyperAgent**
    Yingru Li, Jiawei Xu, Lei Han, Zhi-Quan Luo. *ICML 2024.*
11. **Prior-Dependent Analysis of Posterior Sampling RL with Function Approximation**
    Yingru Li, Zhi-Quan Luo. *AISTATS 2024.*
12. **HyperDQN: A Randomized Exploration Method for Deep Reinforcement Learning**
    Ziniu Li, **Yingru Li**†, Yushun Zhang, Tong Zhang, Zhi-Quan Luo. *ICLR 2022.*
13. **Divergence-Augmented Policy Optimization**
    Qing Wang⋆, **Yingru Li**⋆, Jiechao Xiong, Tong Zhang. *NeurIPS 2019.*

EDUCATION

The Chinese University of Hong Kong (CUHK) *2025*

Ph.D. in Computer Science and Information Engineering.
Advisor: *Zhi-Quan (Tom) Luo*. Committee: Xinyun Chen, Baoxiang Wang, John C.S. Lui, Benjamin Van Roy (Stanford & DeepMind).

Huazhong University of Science and Technology (HUST) *2020*

B.E./M.S. in Computer Science. Ranked 1st/134 overall; 1st/26 in Computer Theory and Software.

SELECTED HONORS

- National Scholarship (China) — highest national academic honor, top 0.2% nationwide, 2018.
- Best Student Paper, IEEE Sensor Array and Multichannel Signal Processing Workshop (SAM), 2024.
- SRIBD Ph.D. Fellowship (Gold Class), 2023; Presidential Ph.D. Fellowship, CUHK, 2019–2023.
- Qiming Star Award — 1 of 5 recipients from 7,112 undergraduates, HUST, 2016.